

Personal information

 **Romain Claret**
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Research Interests

Neuroevolution · Artificial Life · Bio-Inspired · Open-Endedness · Space · Robotics

Education

- 2020 – 2026  **Ph.D. Computer Science** with Prof. Dr. Kilian Stoffel and Prof. Dr. Paul Cotofrei.
Scaling Adaptive Substrate Neuroevolution.
University of Neuchâtel, Neuchâtel, Switzerland
- 2018 – 2020  **M.Sc. HES-SO Software Engineering** with Prof. Dr. Jean Hennebert.
Multi-hop Multi-turns Question-Answering Chatbot using Sub-Knowledge Graphs.
University of Applied Sciences and Arts of Western Switzerland, Lausanne, Switzerland
- 2013 – 2016  **B.Sc. HES-SO Software Engineering** with ing. info. dipl. EPF Marc Schaefer.
Anonymous and Decentralized Browser-based Data Sharing Service.
University of Applied Sciences and Arts of Western Switzerland, Neuchâtel, Switzerland

Grants & Fellowships

- 2023 – 2024  **Doc.Mobility Fellowship (CHF 22'778).** Awarded by swissuniversities and the University of Neuchâtel to fund a 6-month research stay (Sep 2023 – Feb 2024) at the Natural Computing Research & Applications group (NCRA).
University College Dublin, Dublin, Ireland.

Complementary Education

- 2024 – 2025  **GECCO Summer School Computer Science Workshops (6 days).**
RMIT University, Melbourne, Australia / University of Málaga, Málaga, Spain
- 2022 – 2024  **BENEFRI Summer School Computer Science Workshops / Presenting (6 days).**
University of Neuchâtel, Bern, and Fribourg
- 2023  **CUSO Winter School Computer Science Workshops / Presenting (28 Hours).**
University of Neuchâtel, Switzerland
- 2020 – 2024  **CUSO Doctoral Training Computer Science and Skills (112 Hours).**
University of Neuchâtel, University of Geneva, University of Lausanne, Online
- 2020  **Required Doctoral Coursework MSc level Computer Science classes (18 ECTS).**
Machine Learning, Computational Thinking, Programming, Business Analytics.
University of Neuchâtel (UNINE), Neuchâtel, Switzerland
- 2010 – 2012  **Undergraduate Major in Physics (5 ECTS).**
Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland
- 2009 – 2010  **Undergraduate Major in Physics & Mathematics (35 Undergraduate Credits).**
Northeastern University, Boston, Massachusetts, United States of America
- 2009  **Extension School (4 Undergraduate Credits).**
Followed the class of Prof. Dr. Wolfgang Rueckner, Principles of Physics II (PHYS E-1B).
Harvard University, Cambridge, Boston, Massachusetts, United States of America
- 2008  **Summer School (8 Undergraduate Credits).**
Followed the class of Dr. Jonathan Weintraub, Laboratory Electronics (PHYS S-123).
Harvard University, Cambridge, Boston, Massachusetts, United States of America

Experience

- 2023 – 2025  **Visiting Researcher** at UCD Natural Computing Research & Applications Group. (Supervised by Prof. Michael O'Neill).
University College Dublin, Dublin, Ireland.
- 2020 – 2026  **Doctoral Assistant** at the Information Management Institute. Applied Mathematics & Databases for 1st-year Bachelor. (Supervised by Dr. Paul Cotofrei).
University of Neuchâtel, Neuchâtel, Switzerland.
- 2017 – 2018  **IT Independent.** Consulting in Software Engineering and Blockchain. The tasks were to advise technologies, define projects, make prototypes, and conduct workshops.
Geneva area, Lausanne area, and Versicherix AG, Switzerland.
- 2016 – 2017  **Co-Founder** at a Blockchain-based InsurTech Startup. Lead in blockchain and innovation. The tasks were to make software architectures, high-level product schematics, management, workshops, prototypes, and documentation. Administration, fundraising, exhibitions, partnerships, and market studies.
Versicherix AG, Solothurn, Switzerland.
- 2010 – 2015  **Founder** at a Multimedia Streaming Startup. Lead developer and executive. The tasks were to make software architectures, implement prototypes & software, manage digital rights, build the business model, perform market studies, conduct fundraising, create partnerships with film studios, and comply with copyright laws.
Libacy, Neuchâtel, Switzerland.
- 2012  **Summer Internship** in Watchmaking R&D using 3D computer-aided design.
Manufacture Claret, Le Locle, Switzerland
- 2010  **Summer Internship** at Jenks Vestibular Physiology Lab under the supervision of Asst. Prof. Dr. Faisal Karmali. Design an experiment to identify a link between vision and the vestibular system. The tasks were to build the setup in 3D, build the setup by adapting a hydraulic flight simulator, run Matlab simulations, experiment on human subjects, and interpret the results.
Massachusetts Eye & Ear Infirmary, Harvard Medical School, Boston, United States of America
- 2004 – 2006  **Summer Internships** in Watchmaking: 3D computer-aided design (CAD) construction and technical drawing of watch movements (2004), disassembly, reassembly and customization of a mechanical pocket watch (2005), manufacturing, chamfering, and technical control (2006).
Manufacture Claret, Le Locle, Switzerland

Teaching

- 2025  **Computer Science Master's Thesis Supervisor.** Evolutionary Computation Hyperparameter Optimization for Neuroevolution.
University of Neuchâtel, Neuchâtel, Switzerland
- 2020 – 2026  **Teaching Assistant.** Applied Mathematics (Analysis and Linear Algebra). 2 hours lecture and 2 hours QA per week per semester given to first-year bachelors in Economical Science and Data Science. (approx. 150 to 200 students per semester).
University of Neuchâtel, Neuchâtel, Switzerland
-  **Teaching Assistant.** Databases (Modelization, SQL, NoSQL, Visualization). 2 hours QA and student's project management per week during spring semester given to first-year and third-year bachelors in Data Science and Economical Science. (approx. 60 students per semester).
University of Neuchâtel, Neuchâtel, Switzerland
- 2020 – 2022  **Guest Lecturer.** "Demystifying Artificial Intelligence for Health Professionals". 3 hours yearly lecture given to second-year physicians. (approx. 20 students).
University of Geneva, Geneva, Switzerland
- 2021  **Teaching Assistant.** Digital Management and Processing of Multimedia Data (Analyze data, Identify and Formulate issues, Create visualizations). 4 days of project supervision and QA during the fall semester (block course) given to bachelors in Economical Science and Law. (approx. 40 students).
University of Neuchâtel, Neuchâtel, Switzerland

Activities in panels and commissions

- 2025 – 2026  **UniNE.** Member of the University Assembly of the University of Neuchâtel.
- 2023 – 2026  **UniNE.** Committee Member of the Mid-level staff Association (ACINE).
- 2015 – 2017  **HES-SO.** Member of the Participatory Council of the Engineering domain.

Memberships

Scientific & Industry Societies

- 2025 –  **SRA.** Member of the Swiss Robotics Association.
- 2022 –  **IEEE.** Member of the Institute of Electrical and Electronics Engineers.
-  **ACM.** Member of the Association for Computing Machinery.
-  **ISAL.** Member of the International Society for Artificial Life.

Student Societies

- 2023 – 2026  **UniNE.** President of the Computational Thinking Association (Algorithmia).
-  **UniNE.** Treasurer of the Association Suisse Alémanique & Friends (ASA&F).
- 2014 – 2016  **HES-SO.** Head of Communication of the Student Association (REH-SO).
- 2010 – 2012  **EPFL.** Webmaster of the Role-playing Association (JDRpoly).
-  **EPFL.** Committee Member of the Physics Students Association. (Irrotationnels)
-  **EPFL.** Member of the Robotics Association (RoboPoly).
- 2009 – 2010  **Northeastern University.** Member of the Society of Physics Students (SPS).

Skills

Languages	 French Native, English Fluent, Russian Proficient, German Basic
Code & Frameworks	 Python, JAX, Pytorch, NetworkX, Scikit, C++, JS, ...
Machine Learning	 Evolutionary, Deep learning, Edge Models, LLMs API & Local, ...
Computing	 Optimization, HPC, Graphs, Kubernetes, Distributed, Parallel, ...
Dev-Ops	 Git, Docker, Continuous Integration, IaaS, PaaS, SaaS, ...
Miscellaneous	 CAD, Electronics, 3D Printing, Knowledge Bases, $\LaTeX$, Blockchain, ...

Certifications

- 2013  **Entrepreneurship.** Granted by Venturelab, Switzerland.
- 2012  **Inventor Training** (CAD). Granted by Hurni Engineering, Switzerland.
- 2008  **Human Subjects Research.** Granted by CITI program, USA.

Publications

Peer-Reviewed Publications

- [1] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Tensor-Accelerated Eager Multi-Resolution Grids for Evolving Large-Scale Substrates," in *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, San Jose, Costa Rica: ACM, 2026. [DOI: 10.1145/3795101.3805361](#).
- [2] **R. Claret**, A. Gygax, M. O'Neill, P. Cotofrei, and P. Felber, "Early-Stopping Thresholds for ES-HyperNEAT: A Data-Driven Approach from Fitness Dynamics," in *Proceedings of the IEEE Congress on Evolutionary Computation (CEC)*, To appear, IEEE, 2026.

- [3] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Investigating Hyperparameter Optimization and Transferability for ES-HyperNEAT: A TPE Approach," in *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, Melbourne, VIC, Australia: ACM, 2024, pp. 1879–1887. [DOI: 10.1145/3638530.3664144](#).

Manuscripts Under Review

- [4] **R. Claret**, A. Gygax, M. O'Neill, P. Cotofrei, M. Palma Mendes, and P. Felber, "Breaking the Central Bias: Spatially Partitioned Experts for ES-HyperNEAT," Under review at ICPR BIO-inspired Methods for Pattern Recognition Workshop, 2026.
- [5] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "The Quadtree Bottleneck in ES-HyperNEAT: From GPU Barriers to Alternative Discovery Strategies," Under review at ICPR BIO-inspired Methods for Pattern Recognition Workshop, 2026.
- [6] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Multi-Behavioral Evolved Substrates Through Neuromodulation and Activation Selection," Under review at Alife 2026, 2026.
- [7] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Per-Node Activation Function Evolution in Indirectly Encoded Substrates: Solvability, Limits, and Emergent Diversity," Under review at Alife, 2026.
- [8] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Persistent Populations Enable Continual Learning in Evolved Neural Substrates," Under review at Alife, 2026.
- [9] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Per-Node Aggregation Function Evolution in Indirectly Encoded Substrates: Asymmetric Complementarity with Activation," Under review at PPSN, 2026.
- [10] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Co-Evolving Neurotransmitter Vectors in Indirectly Encoded Substrates: A Search Landscape Characterization," Under review at PPSN, 2026.
- [11] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Bio-Inspired Meta-Learning Strategies for Per-Node Activation Function Discovery in Evolved Neural Substrates," Under review at PPSN, 2026.

Conference Abstracts & Posters

- [12] **R. Claret**, *Scaling Neuroevolution for Heterogeneous Tasks using Universal Knowledge Representation*, Poster and ongoing-research presentation, ACM International Conference on Distributed and Event-Based Systems (DEBS), Neuchâtel, Switzerland, 2023.
- [13] F. Karmali, K. Lim, A. Adatia, **R. Claret**, K. Nicoucar, and D. M. Merfeld, *Perceptual Roll-Tilt Thresholds Demonstrate Visual-Vestibular Fusion*, Conference abstract and poster, 40th Annual Meeting of the Society for Neuroscience, San Diego, CA, USA, Nov. 2010.